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ASSIGNMENT- 01

Bachelor of Computer Applications (Hons. AI & ML)
Semester- III

Subject: Database Management System

Topic: Identifying Tables for a Real-World Application

By

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PARANORMAL INVESTIGATION AGENCY MANAGEMENT SYSTEM

1. Name and Purpose of the Application

Name of the Application: Paranormal Investigation Agency Management System

Purpose: A Paranormal Investigation Agency Management System helps an investigation agency organise the cases it takes on — from the moment a client reports unexplained activity at a property, to assigning a team of investigators, checking out equipment, logging the evidence collected on-site, and finally issuing a report with the team's conclusion. The system replaces scattered notebooks and case files with a structured database that connects every client, location, investigator, and piece of evidence to the specific case it belongs to, making it possible to track ongoing investigations, revisit past case history for a location, and manage the agency's equipment and billing accurately

2. List of Required Tables

1. Clients — people who report paranormal activity and request an investigation
2. Investigators — agency staff who carry out investigations
3. Locations — sites/properties being investigated
4. Case_Categories — classification of the type of paranormal activity
5. Equipment — investigation gear owned by the agency
6. Cases — master record of every investigation taken on

7. Case_Investigator — junction table linking cases to the investigators assigned (many-to-many)
8. Case_Equipment — junction table linking cases to the equipment checked out (many-to-many)
9. Evidence — records of everything collected during a case
10. Reports — final write-up and verdict for each case
11. Payments — billing records for investigation services

3. Columns / Attributes

1. Clients Table

Column	Type	Description
client_id	INT (primary Key)	Unique ID for each client
name	VARCHAR	Name of the client
email	VARCHAR	Email address of the client
phone	VARCHAR	Contact number of the client
address	VARCHAR	Client's contact address
registration_date	DATE	Date the client first contacted the agency

Purpose: Stores the contact details of the people or families who report paranormal activity and request an investigation.

2. Investigators Table

Column	Type	Description
investigation_id	INT (Primary Key)	Unique ID for each investigator
name	VARCHAR	Name of the investigator
speciliazation	VARCHAR	EVP Analysis / Photography / Demonology / Skeptic-Debunker
phone	VARCHAR	Contact number of the investigator
email	VARCHAR	Email of the investigator

experience_level	VARCHAR	Trainee / Field Agent / Lead Investigator
date_joined	DATE	Date the investigator joined the agency

Purpose: Maintains records of the team members who visit sites and carry out the actual investigation.

3. Locations Table

Column	Type	Description
location_id	INT (Primary Key)	Unique ID for each location
address	VARCHAR	Address of the site being investigated
location_type	VARCHAR	Residential/ commercial/ historic site/ abandoned building
year_built	INT	Year the structure was built, if known
known_history	VARCHAR	Brief notes on the site's past (previous owners, past incidents)

Purpose: Stores details of the site under investigation, kept separate from Cases since the same location can be investigated more than once if activity continues.

4. Case_Categories Table

Column	Type	Description
Category_id	INT (Primary Key)	Unique ID for each category
category_name	VARCHAR	Residential Haunting / Poltergeist Activity / EVP
		Session / Demonic Activity / Debunking Request
description	VARCHAR	Short description of what this category of activity involves

Purpose: Classifies cases by the type of paranormal activity reported, which helps in assigning the right specialists.

5. Equipment Table

Column	Type	Description
equipment_id	INT (Primary Key)	Unique ID for each equipment item
equipment_name	VARCHAR	EMF Meter / split box/ full-spectrum camera/ digital voice recorder
equipment_type	VARCHAR	Category of the device (Audio / Visual / Sensor)
condition_status	VARCHAR	Working / Under Repair / Retired
purchase_date	DATE	Date the equipment was purchased

Purpose: Keeps an inventory of all investigation gear owned by the agency.

6. Cases Table

Column	Type	Description
Case_id	INT (Primary Key)	Unique ID for each case
Client_id	INT (FK)	References Clients
Location_id	INT (FK)	References Locations
Category_id	INT (FK)	References Case Categories
Date_reported	DATE	Date the client first reported the activity
Investigation_date	DATE	Date the on-site investigation took place

Case_status	VARCHAR	Reported / Under Investigation / Closed / Unresolved
Urgency_level	VARCHAR	Low / Medium / High

Purpose: The central table of the whole system — one row per investigation, tying together who reported it, where it happened, and what kind of activity was involved.

7. Case_Investigator (Junction Table)

Column	Type	Description
Case_id	INT(FK)	References cases
Investigator_id	INT(FK)	References investigators
Role_in_case	VARCHAR	Lead/ support/ equipment handler

Purpose: Resolves the many-to-many relationship between cases and investigators — a case is usually handled by a small team, and one investigator works on many cases over time.

8. Case_Equipment (Junction Table)

Column	Type	Description
Case_id	INT (FK)	References cases
Equipment_id	INT (FK)	References equipment
Checkout_time	DATETIME	When the equipment was taken out for the case
Return_time	DATETIME	When the equipment was returned to storage

Purpose: Resolves the many-to-many relationship between cases and equipment, and doubles as a simple checkout log

— one case can use several devices, and one device gets reused across many cases.

9. Evidence Table

Column	Type	Description
Evidence_id	INT (Primary Key)	Unique ID for each piece of evidence
Case_id	INT (FK)	References Cases
Recorded_by	INT (FK)	References Investigators
Evidence_type	VARCHAR	Photo / Audio Recording / Video / EMF Spike / Temperature Drop / Eyewitness Sighting
Description	VARCHAR	Short note on what was captured or observed
File_reference	VARCHAR	Path or link to the stored photo/audio/video file
timestamp	DATETIME	Date and time the evidence was recorded

Purpose: Logs everything collected during an investigation
— essentially the raw findings of a case, from EVP recordings to eyewitness accounts

10. Reports Table

Column	Type	Description
Report_id	INT (Primary Key)	Unique ID for each report
Case_id	INT (FK)	References Cases
Prepared_by	INT (FK)	References Investigators

Final_verdict	VARCHAR	Confirmed Activity / Debunked / Inconclusive
Summary	VARCHAR	Written summary of the findings and conclusion
Report_date	DATE	Date the report was finalised

Purpose: The final write-up submitted to the client at the end of a case, stating what the team concluded from the evidence gathered.

11. Payments Table

Column	Type	Description
Payment_id	INT (Primary Key)	Unique ID for each payment
Case_id	INT (FK)	References Cases
Amount	DECIMAL	Amount charged for the investigation
Payment_mode	VARCHAR	UPI / Card / Cash / Bank Transfer
Payment_status	VARCHAR	Pending / Paid / Refunded
Payment_date	DATE	Date the payment was made

Purpose: Tracks billing for the investigation service provided to the client.

4. Relationships Between Tables

Relationship	Type	Description
Clients → Cases	One-to-Many	One client can report multiple cases over time

Locations → Cases	One-to-Many	One location can be investigated more than once if activity continues
Case_Categories → Cases	One-to-Many	One category can apply to many cases
Cases ↔ Investigators	Many-to-Many	A case is handled by a team of investigators; one investigator works many cases
Cases ↔ Equipment	One-to-Many	A case can use several pieces of equipment; one device is reused across many cases
Cases → Evidence	One-to-Many	One case can generate many pieces of evidence
Investigators → Evidence	One-to-One	One investigator can record evidence across many cases
Cases → Reports	One-to-One	Each case results in exactly one final report
Investigators → Reports	One-to-Many	One investigator can prepare reports for many cases
Cases → Payments	One-to-One	Each case is billed through exactly one payment record

5. Summary

This design uses 11 tables, each serving a distinct, necessary purpose:

- Master/reference tables: Clients, Investigators, Locations, Case_Categories, Equipment

- Central table: Cases (ties every other table back to a single investigation)
- Junction tables: Case_Investigator, Case_Equipment (resolve the many-to-many relationships)
- Transactional tables: Evidence, Reports, Payments

Every table connects logically through primary and foreign keys, avoiding repeated or unnecessary entities while fully supporting the core operations of a real paranormal investigation agency: taking on a case, assigning a team, tracking equipment, recording evidence, and delivering a final report to the client.